Issue	**1**
Factory ID	**25959**
Date of Issue	**13-Apr-2026**

FA Bhatti Industries

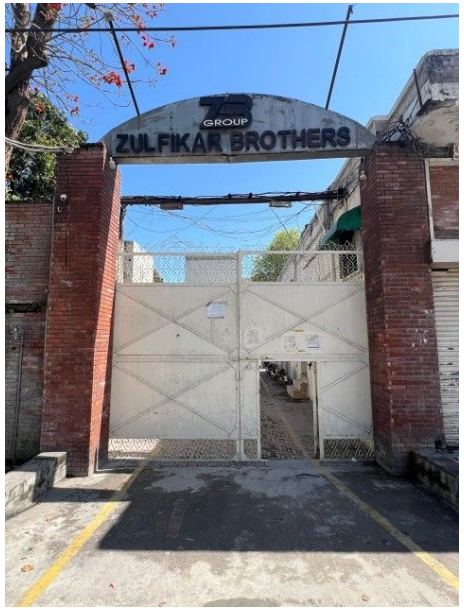
Pakistan ACCORD for Health and Safety in the Garment and Textile Industry

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Date of inspection: 16-Mar-2026

Electrical Safety Inspection Report

Observations & Actions



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Pakistan Accord Initial Safety Inspection Report

About this report:

This report has been prepared by the Health and Safety Accord Pvt Ltd after visiting the factory on 16-03-2026 as part of the inspection and remediation program of the Pakistan Accord on Health and Safety in the Textile and Garment Industry. The purpose of the report is to provide key findings and identify related safety improvements needed at the inspected factories. This inspection report may not identify minor deficiencies, which may be further addressed as part of follow-up inspections.

In addition, while this round of initial inspections covers fire, electrical and structural safety, the Pakistan Accord intends to include inspections on boiler safety and hazardous material in later rounds.

This report will serve to facilitate discussions between the Pakistan Accord, the factories and the responsible brands that source from this facility about the remediation requirements. An implementation schedule shall be developed by the factory in the form of a Corrective Action Plan (CAP) to remediate each of the findings identified in the inspection report, including timelines and cost estimates, for approval by the Chief Safety Officer (CSO).

Applied Standards:

This safety assessment was conducted by the Pakistan Accord with reference to the Pakistan Accord Building Standard.

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1. Executive Summary

On 16-Mar-2026, Azka Riaz and Muhammad Bilal Butt on behalf of the Pakistan Accord carried out an initial electrical inspection for FA Bhatti Industries, a factory at the address and coordinates given on the cover page of this report. The inspection was carried out in accordance with the Pakistan Accord Building standards.

We met with Mr. Azhar Iqbal (General Manager), Mr. Muhammad Afzal (Manager HR), and other staff members of F. A. Bhatti Industries. The assessment was conducted as part of an overall Structural, Fire, and Electrical initial inspection.
The factory operates in a single general shift. The factory employs approximately 150 workers with no female representation.

The complete facility comprises nine main buildings and four ancillary structures. Out of nine, two main buildings are owned by F. A. Bhatti Industries, while the rest of the buildings are owned by ZB (Zulfikar Brothers) Group. During the inspection, the buildings highlighted in blue in Figure 1.1 were surveyed with the site boundary indicated in red.

- **Building 1** is a single-storey (Ground Only) building. The building is used for offices, pattern R&D section (Ground Floor).
- **Building 2** is a two-storey (Ground + 1) building. The building is used for storage, cutting, pressing (Ground Floor) and stitching, checking, packing (First Floor).
- **Building 3** is a single-storey (Ground Only) building. The building is used for mini-accessory store, conference room and offices (Ground Floor).
- **Building 4** is a single storey (Ground Only) building. The building is used for Stitching (Ground Floor).
- **Building 5** is a single storey (Ground Only) building. The building is used for offices, canteen (Ground Floor).
- **Building 6** is a two-storey (Ground + 1) building. The building is used for cutting (Ground floor) and stitching (First Floor).
- **Building 7** is a two-storey (Ground + 1) building. The building is used for cutting (Ground Floor).
- **Building 8** is a two-storey (Ground + 1) building. The building is used for packing, checking (Ground floor) and stitching (First Floor).
- **Building 9** is a two-storey (Ground + 1) building. The building is used for packing, checking, storage (Ground floor) and stitching (First Floor).

Ancillary Structures consist of the following:

- **Toilet Block** is a single storey (Ground Only) structure. The structure is used for toilets.
- **Parking / Storage Shed** is a single storey (Ground only) structural steel shed. The structure is used for parking, waste storage, and steamer.
- **Generator Room** is a single storey (Ground Only) structure. The structure is used for generators, batteries, and panels.
- **Prayer Area** is a single storey (Ground Only) structure used for praying.

A summary of the main concerns in relation to life safety systems identified during the inspection (non-exhaustive) that should be actioned is as follows:

- **Fire Alarm Detection System:** This facility lacks a fire alarm control panel and relies on inadequate standalone detectors and 220V hooters with manual push buttons within its occupied buildings. Additionally, no fire alarm and detection system was observed in other tenant-occupied buildings,

indicating a significant life-safety risk in the event of a fire.
• **Emergency lighting System:** This facility had an emergency lighting system installed; however, coverage was inadequate across its occupied areas. In other tenant-occupied buildings, illuminated directional signage was installed but found non-functional, with no emergency lighting system provided, hindering safe evacuation during power outages and increasing the risk of accidents.
• **Earthing system:** The earthing system was found to be inadequate across the facility. Earthing pits were inaccessible, most distribution boards lacked earthing conductors, and none of the electrical machines were provided with earth connections. The absence of a proper earthing system poses significant safety risks, including electric shock, fire hazards, and potential equipment damage.

We see no reason to suspend operations in the facility due to these concerns, subject to the actions identified being addressed within the specified timeframes.

Our Limitations and Assumptions are also noted at the end of this report.

For detailed analysis of the above, please refer to the full inspection report below, which further details each issue, and also includes photographs. Further actions with associated priorities and timeframes are also detailed with each issue. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

2. Building Extents



Figure 1.1. Site plan (Satellite view) – Red dashed line indicates site boundary, highlighted area (blue) indicates buildings inspected

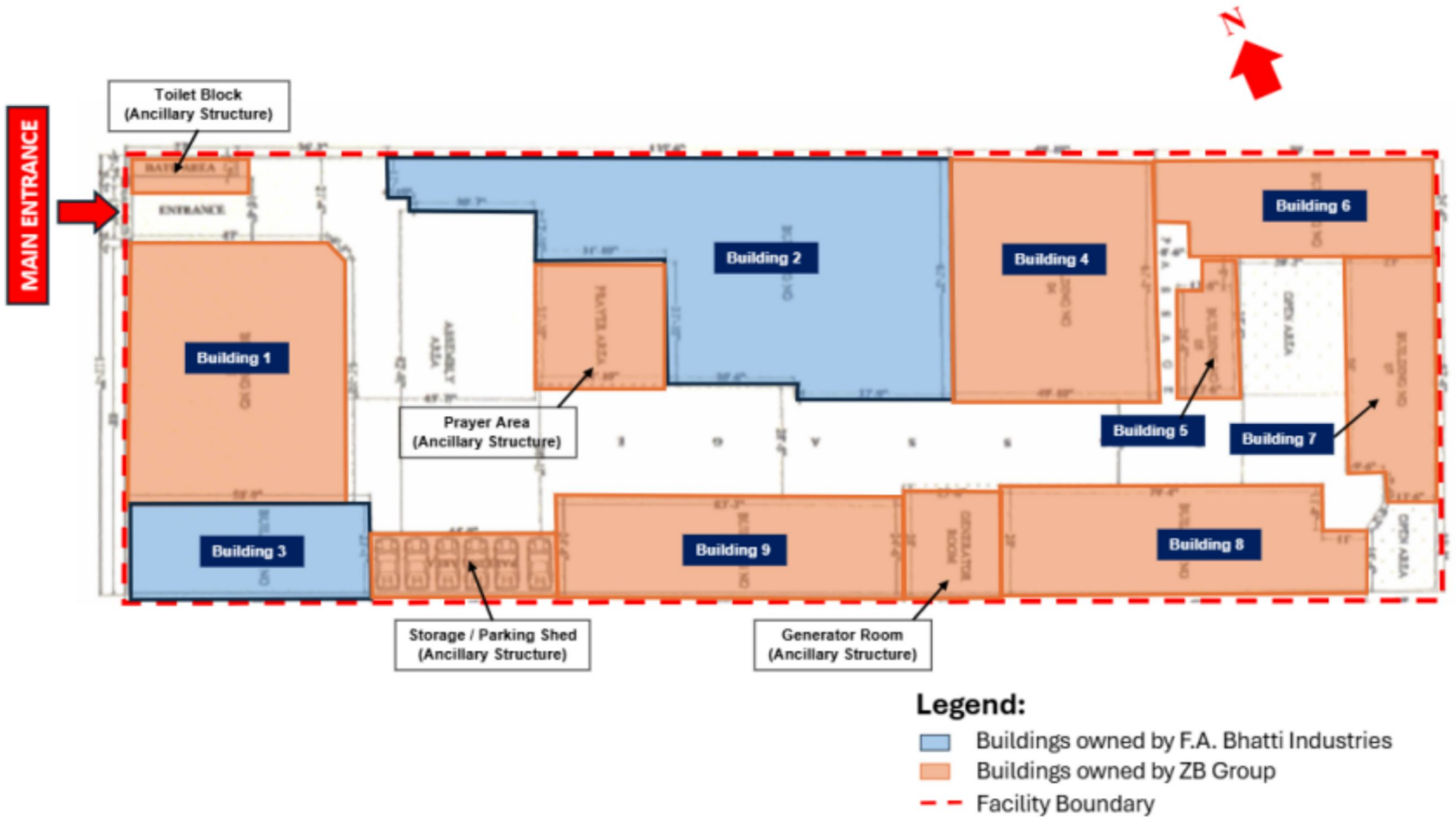


Figure 1.2. Site layout plan of FA Bhatti Industries

2.1 Building Use

The factory comprises nine main buildings and four ancillary structures. The floor-wise activity details are as follows:

1) Building 1
L00 – Ground Floor: Offices and pattern R&D section.
2) Building 2
L00 – Ground Floor: Storage, cutting, and pressing.
L01 – First Floor: Stitching, checking, and packing.
3) Building 3
L00 – Ground Floor: Mini-accessory store, conference room, and offices.
4) Building 4
L00 – Ground Floor: Stitching.
5) Building 5
L00 – Ground Floor: Offices and canteen.
6) Building 6
L00 – Ground Floor: Cutting.
L01 – First Floor: Stitching.
7) Building 7
L00 – Ground Floor: Cutting.
8) Building 8
L00 – Ground Floor: Packing and checking.
L01 – First Floor: Stitching.
9) Building 9
L00 – Ground Floor: Packing, checking, and storage.
L01 – First Floor: Stitching.

Ancillary Structures:
1) Toilet Block
L00 – Ground Floor: Toilets.
2) Parking / Storage Shed
L00 – Ground Floor: Parking, waste storage, and steamer.
3) Generator Room
L00 – Ground Floor: Generators, batteries, and panels.
4) Prayer Area

L00 – Ground Floor: Praying.

2.2 Utility Source

The facility receives a 440 V connection from Gujranwala Electric Power Company (GEPCO) as its primary power source, supplied through a GEPCO-owned step-down transformer installed near the main gate. The incoming power is routed to the Low Tension (LT) panel located in the Generator Room, from where it is distributed throughout the facility via underground cables to various distribution boards.

2.3 Backup source

The facility has three low-voltage diesel generators (220 kVA, 180 kVA, and 1000 kVA) located in the Generator Room, serving as backup power sources. Fire alarm and emergency lighting systems are provided with built-in battery backup.

3. Documentation

This table shows the details of the documentation provided by the factory.

S.NO	Description	Available (Y/N)	Created Date	Status / Remarks
1	Electrical Layout Drawing	No		
2	Single Line Diagrams	No		
3	Fire Alarm Drawing	No		
4	Earth Test Report	No		
5	Thermographic Test Report	No		
6	Insulation Resistance Test Report	No		
7	LPS Risk Assessments	No		
8	Periodic Maintenance Record	No		
9	Transformer Oil Testing Report	N/A		

4. Electrical Safety Measures

During our electrical site inspection of FA Bhatti Industries, a number of electrical issues were identified.

It is essential to address these electrical safety issues promptly to enhance safety and compliance within FA Bhatti Industries. Each issue should be carefully evaluated, and the recommendations provided in this report should be followed diligently to rectify these concerns. The safety and well-being of the factory's employees depend on these corrective actions. Where issues are identified in the report, they are grouped by topic, not in order of importance. Please note that these actions should be completed as soon as practically possible and certainly within the timeframe noted.

Recommendations in this report are prioritized as follows:

Immediate: These recommendations address safety and compliance concerns that require immediate attention.

Medium term (within approx. 6-weeks): Recommendations of high importance but needing slightly more time to implement.

Longer term (within approx. 6-months): Important recommendations that require a longer period to fully implement or that enhance long-term safety and efficiency.

All survey, testing, and design activities specified in this report shall be performed by a qualified electrical engineer. The recommended installation and modification tasks should be executed by a licensed professional electrician.

Before commencing any installation and/or modification works, the proposed design must be submitted to the Pakistan Accord for approval.

Upon completion of the works, the necessary evidence required to verify that the tasks have been completed to the agreed standards will be determined during the development of the Corrective Action Plan (CAP).

5. Remediation Support

5.1 Engineering Support

Some items in the Corrective Action Plan (CAP) will require remediation by a qualified engineer. Please refer to the Remediation Plan for guidance on which tasks can be completed using internal resources and which require engineering support. F-202603-064364

Timeline	Remediation Plan
Medium Term CAP-0075835	Factories should begin remediation of CAP items such as cleaning, cable organization, and basic corrections using internal resources. For systems or design-related remediation, engagement of a qualified Electrical Engineer is required.

6. Diagrams & Drawings

6.1 Sub-station room layout drg

This section refers to the electrical rooms within the facility.

During the inspection conducted, it was observed that there were no substation room layout drawings available for the factory.

There are no visible indications of substation equipment locations or pathways for maintenance and emergency access.

The absence of substation room layout drawings poses significant risks to life and fire safety within the factory premises. Without these critical documents, emergency responders and maintenance personnel may face difficulties in locating and isolating electrical faults during emergencies. This delay can result in prolonged exposure to electrical hazards, increasing the risk of electrical fires and endangering lives. Furthermore, the lack of layout drawings complicates routine maintenance, potentially leading to undetected faults and increased likelihood of equipment failure. F-202603-064249

Timeline	Remediation Plan
Medium Term CAP-0075545	Develop detailed substation room layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). Ensure that all critical equipment and pathways are accurately documented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075546	Regularly update and review the substation room layout drawings to reflect any changes in equipment or configuration. Provide training sessions for maintenance personnel and emergency responders on how to interpret and utilize the layout drawings.

6.2 Single Line Diagram (SLD)

This section refers to the entire facility.

It was found that the electrical single line diagrams (SLDs) for the factory were missing.

This deficiency was noted in all electrical rooms and areas where SLDs should be prominently displayed.

The factory lacks documented diagrams that illustrate the electrical distribution system, including the connection and configuration of all electrical components.

Single line diagrams are crucial for understanding the electrical distribution network and for troubleshooting issues effectively. Without these diagrams, maintenance personnel and emergency responders lack critical information necessary for isolating faults and managing power during emergencies. This can lead to delays in addressing electrical issues, increasing the risk of electrical fires and potential harm to personnel. Moreover, the lack of SLDs complicates routine maintenance and inspection processes, potentially resulting in undetected electrical faults and system inefficiencies.F-202603-064254

Timeline	Remediation Plan
Medium Term CAP-0075543	Develop comprehensive electrical single line diagrams for the entire facility in accordance with the requirements outlined in Section 514.9.1 of BS7671 18th Edition + A2 (2022). Ensure all components and connections are accurately represented. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075544	Implement a regular review and update process for the single line diagrams to reflect any changes in the electrical system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the diagrams.

6.3 Electrical layout drawing

This section refers to the entire facility.

During the survey, it was noted that the factory lacked electrical layout drawings - including small power layouts, lighting layouts, and containment layouts.

This absence was consistent across all areas where such documentation should be available.

These drawings are vital for understanding the placement and connections of electrical systems, which is crucial for identifying and mitigating potential hazards. In emergencies, the lack of detailed layouts can delay the identification and isolation of electrical faults, increasing the risk of fire and electrical accidents. Additionally, without proper layout drawings, routine inspections and maintenance may overlook critical issues, potentially leading to system failures and safety breaches.F-202603-064248

Timeline	Remediation Plan
Medium Term CAP-0075547	Develop comprehensive electrical layout drawings in accordance with the requirements outlined in Sections 132.12 and 132.13 of BS7671 18th Edition + A2 (2022). To avoid drawings becoming overcrowded, separate drawings shall be prepared for each electrical system. Retain the drawings along with all O&M information for the facility.
Longer Term CAP-0075548	Implement a regular review and update process for the electrical layout drawings to reflect any changes in the system. Conduct training sessions for maintenance personnel and emergency responders on interpreting and using the layouts.

7. DBs & Elec Protection Systems

7.1 Circuit breakers

This section refers to the entire facility.

During the survey, the circuit breakers installed throughout the facility, inside the electrical panels and distribution boards, were inspected.

i. It was observed that the circuit breaker was installed outside building 1 without proper fixing supports and protective enclosures. This inadequate fixing could lead to poor connections, arcing, short circuits, or other electrical faults.

ii. It was observed that multiple circuit breakers had ratings that were not appropriate for the connected cable sizes.
a. In the LT panel, a 500A circuit breaker was connected to 75 mm^2 PVC cables, and 175A &150A circuit breakers were connected to 25 mm^2 PVC cables.
b. In the stitching distribution board of building 3, a 125A circuit breaker was connected to 2.5 mm^2 PVC cables.

Specifically, some circuit breakers had higher ratings than the cables could safely handle, which poses a significant risk of overheating and potential fire hazards. Conversely, other circuit breakers had lower ratings than necessary, resulting in frequent and unnecessary tripping. To ensure both safety and operational efficiency, corrective action is required to align the circuit breaker ratings with the cable capacities. This adjustment will mitigate the risks associated with overheating and improve the reliability of the electrical system. F-202603-064239

Timeline	Remediation Plan
Medium Term CAP-0075818	i. Properly mount and relocate the circuit breaker to indoor or sheltered areas. If relocation is not feasible, install weatherproof enclosures with a suitable IP rating for outdoor conditions, as per BS 7671 Clause 512.2.

Medium Term CAP-0075819	ii. Evaluate the breaker ratings and cable sizes for all circuits in the facility according to Section 433.1.1 and Appendix 4 of BS 7671 18th Edition + A2 (2022). Ensure that the breaker rating does not exceed the cable's current-carrying capacity. If the breaker has adjustable settings, they must be configured to allow current within the cable's ampacity after applying derating factors.



Figure 2.1 - 125A circuit breaker connected to 2.5 mm² PVC cables, stitching distribution board, building 3



Figure 2.2 - 175A &150A circuit breakers connected to 25 mm² PVC cables, LT panel



Figure 2.3 - 500A circuit breaker connected to 75 mm² PVC cables, LT panel



Figure 2.4 - Non-weather protected and improperly fixed circuit breaker installed outside building 1

7.2 Hot Spots

This relates to the entire facility.

During the inspection, a thermographic survey could not be effectively conducted due to insufficient electrical load conditions across the facility; therefore, the observations may not represent performance under full load.

No hotspots were observed at the time of inspection; however, the overall electrical load was inadequate to perform a comprehensive thermographic assessment. Additionally, no thermographic survey reports were available on-site for review.

Without proper thermography testing under adequate load conditions, potential issues such as loose connections, overloaded circuits, faulty components, improper cable sizing, or insulation degradation may remain undetected, increasing the risk of overheating and fire incidents.

It is recommended that the facility conduct a complete thermographic survey under normal operating load conditions and maintain proper records to ensure early detection of faults and continued safe operation of the electrical system. F-202603-064241

Timeline	Remediation Plan
Longer Term CAP-0075817	Regular thermographic inspections should be incorporated into the maintenance schedule to proactively manage and mitigate the risks associated with overheating. Submit the thermographic test report to the Pakistan Accord for review.

7.3 Earthing system

This relates to the entire facility.

During the inspection of the facility's electrical panels, distribution boards, and machinery, several observations with the earthing system were identified.

i. No earthing system design documentation was provided by the factory, indicating a lack of verification of the earthing system’s compliance and ongoing performance in accordance with relevant standards.

ii. The earth pits were inaccessible and unsuitably located. This issue was observed in the generator room.

iii. The earthing system was absent in multiple locations. Several distribution boards lacked earthing cables, and many electrical machines did not have proper earth connections. This issue was specifically observed in the distribution boards of buildings 1 and 9, as well as with the cutting and stitching machines in buildings 3 and 9.

iv. Equipotential bonding for the cable trays, metallic panel covers, generator body, and frames was missing.

v. The earth pit testing and maintenance records were not provided.

The lack of a proper earthing system throughout the facility poses substantial safety risks, such as electric shocks, fires, and equipment damage. Implementing appropriate earthing measures is crucial to prevent hazardous voltage levels and ensure operational safety. F-202603-064240

Timeline	Remediation Plan
Medium Term CAP-0075823	i. Design and evaluate the facility's earthing system, and develop layout drawings in accordance with BS 7671 18th Edition + A2 (2022). Submit the earthing layout drawings to the Pakistan Accord for review.
Medium Term CAP-0075820	ii. Relocate and redesign earth pits to make them easily accessible for maintenance and inspection, complying with BS 7671 18th Edition + A2 (2022) Chapter 54.
Medium Term CAP-0075821	iii. Install earthing systems where absent, ensuring compliance with BS 7671 18th Edition + A2 (2022) Chapter 54. Fit all electrical boards with the required earthing cables. Ensure that earthing conductors are properly connected and comply with BS 7671 18th Edition + A2 (2022) Section 543.1.

Medium Term CAP-0075822	iv. Implement equipotential bonding for all cable trays to comply with BS 7671 Regulation 411.3.1.2. Ensure all exposed conductive parts are bonded to the main earthing terminal.
Medium Term CAP-0076099	v. Implement periodic earth testing per BS 7671 Regulation 643.7.1 to verify grounding effectiveness and submit testing record to Pakistan Accord for review.
Longer Term CAP-0075824	Establish a schedule for regular maintenance and inspection of the earthing system and protective devices as per BS 7671 Regulation 651.2. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on proper earthing practices and the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.5 - Earth connection missing on the first floor distribution board of building 9



Figure 2.6 - Earthing pit inaccessible and unsuitably located in the generator room



Figure 2.7 - Earth connection missing for the stitching machine, building 3



Figure 2.8 - Earth connection missing for the cutting machine, building 3



Figure 2.9 - Equipotential bonding missing for the generator body and frames

7.4 Panel/cabling/busbar

This relates to the entire facility.

The visual site inspection included an examination of the electrical panels, cables, and busbars throughout the facility.

i. It was observed that electrical components were not suitable for the environmental conditions in which they were installed. This was observed with the cable termination box installed at the main gate entrance and electrical enclosures installed outside buildings 4 and 8.

ii. There were visible signs of damage to panels and cables, including broken panel doors, faulty phase indication lights, and non-operational voltmeters and ammeters. This was observed with the LT panel and glove distribution board of building 2.

iii. Corrosion was observed on a distribution board in building 5, indicating potential exposure to moisture or other corrosive elements.

iv. An electrical component was observed without proper mounting outside the building 8 electrical enclosure, posing a risk of loose connections, mechanical damage, and electrical faults.

v. Phase separators were missing in LT panels, posing the risk of accidental contact between phases.

vi. Inadequate cable dressing was observed in most of the distribution boards, with conductors not properly arranged, secured, or terminated neatly. This resulted in untidy wiring, poor segregation between circuits, and unnecessary strain on terminations, which can lead to loose connections, overheating, and difficulty during inspection or fault-finding. This issue was specifically observed with the LT panel and changeover panels in the generator room, the glove distribution board of building 2, the stitching distribution board of building 9, as well as the electrical enclosure installed on the first floor of building 8.

The utilization of electrical equipment that is either unsuitable for its operating environment or physically damaged poses significant safety risks. These risks include potential electrical failures, increased likelihood of short circuits, and the heightened risk of fire and electrical shock. F-202603-064242

Timeline	Remediation Plan
Medium Term CAP-0075828	i. Identify and replace electrical panels, cables, and busbars that are not suitable for the environmental conditions in which they are installed in accordance with Section 512.2 of BS 7671 18th Edition + A2 (2022).

Medium Term CAP-0075829	ii. Inspect and repair or replace all damaged panel doors, locks, and cables to ensure the integrity of the electrical system in accordance with Section 522 of BS7671 18th Edition + A2 (2022). Replace the defective voltmeters, ammeters, and indicator lights with new and reliable components.
Medium Term CAP-0075830	iii. Replace or refurbish corroded electrical panels to prevent further degradation and potential safety hazards. Implement protective measures such as installing weatherproof enclosures or using corrosion-resistant materials.
Medium Term CAP-0075831	iv. Properly secure and mount all electrical components within panels using appropriate mounting arrangements to prevent loose connections, mechanical damage, and potential electrical faults.
Medium Term CAP-0075832	v. Phase separators should be installed between adjacent live conductors of different phases within distribution boards to ensure safe segregation and protection against direct contact, in accordance with BS 7671 Regulation 416.2.
Medium Term CAP-0075833	vi. Ensure proper cable management by organizing and securing cables within the panels using cable ties, clips, or ducting. Verify that all components and connections are securely mounted, neatly arranged, and labelled to improve safety, ease of maintenance, and compliance with electrical standards such as BS 7671.
Longer Term CAP-0075834	Establish a schedule for regular maintenance and inspection of the panel, cabling and busbars. Document all inspections and maintenance activities to ensure ongoing compliance. Provide training to facility maintenance personnel on the importance of maintaining electrical safety standards in line with BS 7671 requirements.



Figure 2.10 - Corrosion observed on a distribution board in building 5



Figure 2.11 - Non-IP rated cable termination box installed at main gate entrance



Figure 2.12 - Inadequate cable dressing observed in the stitching distribution board, building 9



Figure 2.13 - Non-operational phase indication lights and meters installed at LT panel



Figure 2.14 - Inadequate cable dressing and door glass broken in glove distribution board, building 2



Figure 2.16 - Phase separators missing in the main LT panel



Figure 2.15 - Non-IP rated electrical enclosure with unmounted electrical component found outside building 8

7.5 Loop Connection

This relates to the entire facility.

During the inspection, loop connections were identified within the electrical panels. This issue was specifically observed in the glove distribution board of building 2, the cutting and stitching distribution boards of building 3, the stitching distribution boards of building 8, as well as in the electrical enclosure installed outside building 4.

In these loop connections, one circuit breaker is energized by the live part of another circuit breaker, creating loops within the panel instead of using a busbar system.

Avoiding loop connections is crucial due to several significant risks. Loop connections can complicate maintenance activities, making it difficult to trace and isolate circuits. These connections can lead to increased resistance and overheating, potentially damaging components. Loop connections raise the risk of accidental phase mixing, which can result in short circuits. The above issues compromise personal safety and significantly increase the risk of fire.F-202603-064244

Timeline	Remediation Plan
Medium Term CAP-0075815	All electrical panels shall be modified to eliminate loop connections. A busbar system within the electrical panels shall be utilized to provide each circuit with the required live, neutral, and earth connections.



Figure 2.17 - Loop connection found in the cutting distribution board of building 3



Figure 2.18 - Loop connections found in the stitching distribution board of building 3



Figure 2.19 - Loop connections found in the enclosure installed outside building 4



Figure 2.20 - Loop connections found in the stitching distribution board of building 8

7.6 Multiple Connections

This relates to the whole facility.

During our inspection, multilooping was observed in a number of electrical panels across the facility. This issue was specifically observed with the gate office distribution board of building 1, the cutting and stitching distribution boards of building 3, and the stitching distribution boards of building 9.

In these cases, multiple circuits were being protected by a single circuit breaker.

Multilooping can result in improper distribution of electrical current, leading to voltage drops and potential electrical hazards. To mitigate these risks and ensure the safety and reliability of the electrical system, it is essential to adhere to proper installation practices. This includes using appropriate dedicated circuit breakers for each circuit.F-202603-064243

Timeline	Remediation Plan
Medium Term CAP-0075816	The electrical installation throughout the facility shall be modified to ensure each circuit is protected by a dedicated circuit breaker of a suitable type and rating for each individual circuit in accordance with Section 430 of BS 7671 18th Edition + A2 (2022).

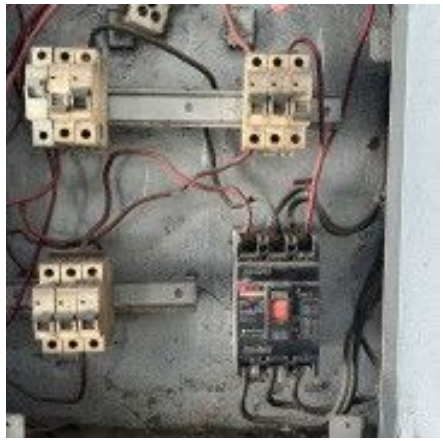


Figure 2.21 - Multiple connections found in the gate office distribution board of building 1



Figure 2.22 - Multiple connections found in the stitching distribution board of building 9

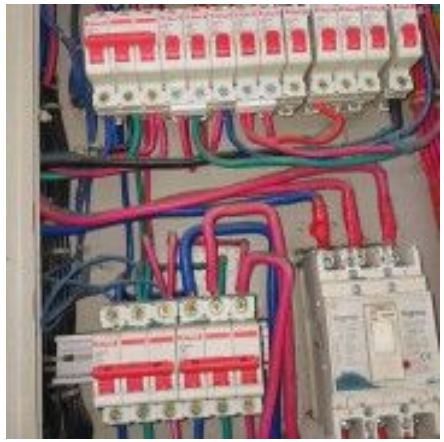


Figure 2.23 - Multiple connections found in the cutting distribution board of building 3



Figure 2.24 - Multiple connections found in the stitching distribution board of building 3

7.7 Openings

This relates to the entire facility of the factory.

During the visual inspection, the electrical panels and distribution boards were examined for unsealed openings to ensure compliance with the IP rating requirements.

The inspection revealed several observations:

i. The front doors and panel covers were found broken or missing. This issue was specifically observed in the stitching distribution board of building 4 and in electrical enclosures installed on the first floor of building 8 and outside building 4.

ii. Unsealed openings were observed in the panels and distribution boards. This issue was specifically observed in the changeover panel of the generator room, the gate office distribution board of building 1, and the gloves distribution board of building 2.

These deficiencies pose multiple hazards. Unsealed openings and missing panels expose live electrical components, significantly increasing the risk of accidental contact and electric shock to personnel working nearby. The unsealed openings allow dust, debris, and moisture to enter, which can lead to short circuits and increase the risk of fire. The absence of proper sealing compromises the IP rating of the panels, which is crucial for ensuring safety and protection against environmental factors.F-202603-064206

Timeline	Remediation Plan
Immediate CAP-0075521	i. Temporarily cover all open electrical enclosures and distribution boards, especially those installed outdoors, using non-combustible protective covers to prevent accidental contact, dust ingress, and moisture exposure until proper panel doors or covers are installed.
Medium Term CAP-0075811	ii. Corrective actions are required to seal all openings and replace missing panel doors and covers to restore the integrity and safety of the electrical systems. The electrical system shall ensure compliance with Sections 614.2.1, 412.2.2, and 512.2 of BS 7671 18th Edition + A2 (2022).
Longer Term CAP-0075812	Conduct regular maintenance and maintain comprehensive records for all electrical panels to ensure there are no openings and that the IP ratings of the panels remain uncompromised.



Figure 2.25 - Unsealed openings observed in the gate office distribution board of building 1



Figure 2.26 - Front cover missing in electrical enclosure on the first floor of building 8



Figure 2.27 - Front door missing in the stitching distribution board of building 4



Figure 2.28 - Unsealed openings observed in the gloves distribution board of building 2

7.8 Adequate accessibility

This relates to the entire facility.

During the visit, the electrical panels and distribution boards were examined for adequate accessibility.

At numerous locations, it was found that the electrical panels do not possess the required space for inspection or easy access to the panels and their components. This issue was specifically observed in building 5.

Limited accessibility to distribution panels and machine panels hampers effective maintenance and inspection procedures. This can potentially cause delays in addressing emergencies and troubleshooting electrical problems. These accessibility challenges could compromise the operational reliability and safety of the premises, as prompt maintenance and emergency responses are hindered. F-202603-064246

Timeline	Remediation Plan
Medium Term CAP-0075810	Electrical equipment shall be arranged to provide sufficient space for the initial installation and later replacement of individual items of electrical equipment. Sufficient allowance shall also be made for sufficient space for the initial installation and later replacement of individual items of electrical equipment in accordance with Section 132.12 of BS7671 18th Edition + A2 (2022).



Figure 2.29 - Inadequate accessibility for the distribution board in building 5

7.9 Lint & dust inside panel board

This relates to the entire facility.

During the inspection, the cleanliness of electrical panels, distribution boards, and electrical containment was assessed.

The inspection revealed a significant accumulation of lint and dust in most of the distribution boards, panels, and trunking systems.

The presence of lint and dust inside distribution boards poses several hazards. Accumulated lint and dust can act as fuel for fires if exposed to electrical currents or sparks, significantly increasing the risk of electrical fires. The cooling capacity of electrical components can also be impaired by the presence of dust and lint, leading to overheating. This issue was specifically observed in the power factor improvement panel in the generator room and in distribution boards installed in buildings 1, 3, and 4.

Additionally, the accumulation of dust and lint can obstruct proper operation and maintenance, thereby impairing the functionality of the electrical components.F-202603-064247

Timeline	Remediation Plan
Medium Term CAP-0075802	All electrical panels and trunking runs shall be inspected and cleaned of any dust or lint in accordance with Section 522.4.2 of BS7671 18th Edition + A2 (2022).
Longer Term CAP-0075803	A regular maintenance procedure shall be established to ensure routine inspection and cleaning of all electrical panels and trunking runs. This will prevent the accumulation of lint and dust, thereby maintaining the safety and efficiency of the electrical systems.



Figure 2.30 - Lint and dust found in the gate office distribution board, building 1



Figure 2.31 - Lint and dust found in the stitching distribution board, building 3



Figure 2.32 - Lint and dust found in the power factor improvement panel, generator room



Figure 2.33 - Lint and dust found in the stitching distribution board, building 4

7.10 Lightning Protection System (LPS)

During the inspection, the lightning protection strategy of the factory was assessed.

It was observed that no lightning protection system was installed, and there was no risk assessment provided to confirm whether it was not required.

Lightning strikes can cause significant damage to buildings, primarily through the ignition of fires. When lightning strikes a structure, it can cause an electrical surge that ignites flammable materials. Without a proper lightning protection system (LPS), the energy from a lightning strike can travel through the building's electrical and plumbing systems, creating sparks and starting fires. F-202603-064250

Timeline	Remediation Plan
Medium Term CAP-0075795	Engage a qualified lightning protection specialist to perform a comprehensive risk assessment according to BS EN 62305 (Protection against Lightning) standard. This assessment will determine the necessity of a lightning protection system (LPS) based on factors such as building height, location, occupancy, and the presence of sensitive electronic equipment.
Longer Term CAP-0075796	If the risk assessment concludes that an LPS is necessary, proceed with the design and installation of an appropriate system in compliance with relevant standards such as BS EN 62305 (Protection against Lightning). Submit the design to Pakistan Accord for review.

8. Fire Alarm & Detection System (FADS)

8.1 Non-functional or incomplete

This section refers to the entire facility.

During the inspection, the Fire Alarm and Detection System (FADS) for the facility was evaluated.

The facility lacks a fire alarm control panel and relies on inadequate standalone detectors and 220V hooters with manual push buttons within its occupied buildings. Additionally, no fire alarm and detection system was observed in other tenant-occupied buildings across the facility, indicating a significant life-safety risk in the event of a fire.

The use of standalone detectors and alarms, along with the absence of an appropriate control panel, constitutes non-compliance with the Building Code of Pakistan Fire Safety Provisions, compromising the system's effectiveness and reliability.

A non-functional/incomplete FADS significantly increases the risk of worker injury or fatality, substantial property damage, operational disruptions, legal liabilities, reputational damage, and environmental harm. A delayed response during a fire event can significantly increase the impact on personnel, goods, and property.F-202603-064205

Timeline	Remediation Plan
Immediate CAP-0075520	Implement an alert mechanism to promptly notify all personnel in case of an emergency and ensure staff are trained to respond appropriately.
Medium Term CAP-0075549	i. Submit design documentation for the new replacement of FADS system to Pakistan Accord for review before purchasing any new equipment. Design documentation to include drawings, technical data sheets, and product certifications of all listed equipment to show full compliance with the Fire Alarm and Detection System (FADS) with NFPA 72.
Longer Term CAP-0075550	ii. Submit documentation to the Pakistan Accord showing that an L/C has been opened for import of any components required for the FADS design submitted earlier.

Longer Term CAP-0075551	iii. Submit documentation to the Pakistan Accord demonstrating that all materials, both imported and locally procured, have been purchased and are available on-site for installation.
Longer Term CAP-0075552	iv. Once installation of the FADS system is complete, the factory is to submit full "as-built" documentation of the system to the Pakistan Accord for review.
Longer Term CAP-0075553	v. Factory to advise Pakistan Accord when FADS installation is fully completed and ready for review. Pakistan Accord will then arrange a verification inspection to ensure that the installation fully complies with the previously reviewed design documentation.
Longer Term CAP-0075554	vi. Provide full testing and commissioning (T&C) certification from the FADS system installer and make provision for attendance of the system installer, factory technical personnel, and Pakistan Accord technical team to witness a final T&C demonstration verification on site. On successful completion of this T&C verification inspection, remediation of the FADS system will be complete.



Figure 2.34 - Fire alarm and detection missing on the first floor of building 9



Figure 2.35 - Manual fire alarm hooter installed on the ground floor of building 3



Figure 2.36 - Standalone smoke alarm installed on the ground floor of building 3



Figure 2.37 - Fire alarm and detection coverage missing in the generator room



Figure 2.38 - Fire alarm and detection coverage missing in building 2

9. Emergency Lighting

9.1 Provided but non-functional

The section refers to the entire facility.

During the inspection, it was observed that emergency exit signs were provided but not functional in a number of locations without any nearby emergency lights. This issue was specifically observed at the main entrance of building 2 and on the first floor of building 6.

Functional emergency lighting is critical for providing illumination in the event of a power failure or other emergencies, ensuring that workers can safely evacuate the facility.F-202603-064251

Timeline	Remediation Plan
Medium Term CAP-0075630	Conduct a thorough inspection of all emergency lighting units throughout the facility to identify specific locations and units that are non-functional. This involves checking both the lights and the power sources, including batteries and wiring connections. Repair any faulty emergency lighting units immediately. If repair is not feasible, replace the defective units with new, compliant emergency lighting fixtures to ensure they provide adequate illumination during an emergency.
Longer Term CAP-0075631	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.39 - Non-functional emergency exit sign found on the first floor of building 6



Figure 2.40 - Non-functional emergency exit sign found at the main entrance of building 2

9.2 Provided but insufficient

The section refers to the entire facility.

During the inspection, the emergency lighting arrangement throughout the facility was assessed.

The following observations were made:

i. Several areas lacked sufficient emergency lighting and illuminated exit sign coverage. This issue was observed in the generator room, building 2, and the ground-floor store of building 3 within FA Bhatti Industries, as well as across all other tenant-occupied buildings. Additionally, the layouts for emergency lighting were missing. The absence of emergency lighting layouts throughout the facility raises concerns regarding the adequacy of illumination provided during power failures or emergencies.

ii. Multiple emergency luminaires indicate the need for maintenance. This issue was specifically observed on the ground floor of building 8.

iii. External emergency lighting was not provided, particularly in the assembly area and external emergency stairs, preventing people from reaching a place of safety outside the facility during nighttime evacuations.

Adequate and functional emergency lighting throughout the facility is crucial for providing necessary illumination during power failures or other emergencies, ensuring the safe evacuation of workers.F-202603-064252

Timeline	Remediation Plan
Medium Term CAP-0075660	i. Conduct a comprehensive survey to identify areas lacking adequate emergency lighting. Prepare and submit the emergency light design and layouts to the Pakistan Accord for review. Install additional emergency luminaires and exit directional signs in these areas to meet the coverage requirements specified in the Building Code of Pakistan Fire Safety Provisions 2016, ensuring sufficient illumination for safe evacuation.
Medium Term CAP-0075661	ii. Perform maintenance on all emergency luminaires indicating issues, including cleaning, bulb replacement, and verification of electrical connections.
Medium Term CAP-0075662	iii. Install external emergency luminaires to enable safe evacuation during hours of darkness, all the way to a safe place outside the building(s).

Longer Term CAP-0075664	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.



Figure 2.41 - External emergency lights not provided in the assembly area



Figure 2.42 - Emergency light and exit sign missing on the first floor of building 9



Figure 2.43 - Emergency light and exit sign not provided in the store on the ground floor of building 3



Figure 2.44 - Non-illuminated exit sign without nearby emergency light in building 4



Figure 2.45 - Emergency lighting missing in the generator room



Figure 2.46 - Emergency exit sign needs maintenance on the ground floor of building 8

9.3 Independent power source

The section refers to the entire facility.

During the inspection, it was observed that the emergency lighting within the facility was not connected in a manner that would ensure automatic activation upon the loss of power to the general lighting in various spaces.

Proper circuitry is crucial to ensure that emergency lighting is automatically activated when general lighting fails. This automatic activation is essential for maintaining sufficient illumination, allowing for the safe evacuation of occupants and preventing panic during power outages. Further the emergency lights and exit signs should be reconfigured to be connected on a dedicated single circuit, which must not include any other load except for the emergency lighting system. According to the Building Code of Pakistan Fire Safety Provisions 2016, emergency lighting systems must be designed to provide immediate illumination in the event of power loss to ensure a clear and safe path for evacuation.

Ensuring that the emergency lighting is integrated with the facility's power system to activate automatically during power failures is critical for life safety. This integration helps prevent hazardous conditions that can arise in the dark, such as trips, falls, and confusion, thereby facilitating orderly and efficient evacuation during emergencies.F-202603-064253

Timeline	Remediation Plan
Medium Term CAP-0075586	Reconfigure the emergency lighting circuitry to ensure that emergency lights activate immediately upon failure of general lighting power as required by the Building Code of Pakistan Fire Safety Provisions 2016. Emergency lighting shall be powered from a source independent of that supplying the normal lighting.
Longer Term CAP-0075587	Establish a regular maintenance and testing schedule to routinely check the functionality of emergency lighting systems. This should include periodic tests, inspections, and prompt maintenance actions to address any issues. Maintain detailed records of all inspections, tests, repairs, and maintenance activities. Train facility maintenance staff on the procedures for inspecting and maintaining emergency lighting systems to ensure ongoing reliability and compliance with safety standards.

10. Generator, Transformer & Sub-Station Room

10.1 Ventilation and illumination

This section refers to the transformer room, sub-station rooms, and generator room.

An assessment of the existing ventilation system was conducted, alongside an evaluation of the lighting setup in various parts of the room, to determine their compliance with recommended standards.

The following observations were made:

i. It was observed that the electrical supply for the lighting and exhaust fan was disconnected in the LT panels and generator area. One light bulb connected to the UPS was present; however, the UPS was out of order.

ii. Several areas within the generator room were poorly illuminated, with lighting levels below those required for safe operation and maintenance.

iii. The ventilation system was inadequate, with insufficient airflow resulting in heat accumulation within the generator room.

BS7671 18th Edition + A2 (2022) specifies the need for ventilation to prevent overheating and ensure the safe operation of electrical installations. The standard also emphasizes that adequate lighting must be provided to ensure visibility for safe operation and maintenance. Proper lighting levels are crucial to avoid accidents and to ensure that any issues with the electrical installation can be promptly identified and addressed. F-202603-064259

Timeline	Remediation Plan
Immediate CAP-0075605	i. Immediately restore electrical supply to lighting and exhaust fans in the LT panels and generator area, and ensure temporary illumination and ventilation are provided until permanent connections are reinstated.
Medium Term CAP-0075606	ii. Add more lighting fixtures in poorly lit areas to ensure even distribution of light throughout the room. Focus on areas that require the highest visibility, such as around equipment and control panels.

Medium Term CAP-0075607	iii. Hire a qualified engineer to reassess the ventilation system to ensure it meets safety and performance standards. If the assessment concludes that the current system is inadequate, redesign the ventilation layout to enhance the distribution of airflow, ensuring that critical areas around the HT panel and transformer receive adequate ventilation. This may involve repositioning vents or adding ductwork as necessary. Submit the ventilation assessment report and the proposed design to Pakistan Accord for review.
Longer Term CAP-0075608	Establish regular maintenance and testing of the ventilation system and luminaires within the spaces to ensure adequate performance.



Figure 2.47 - Inadequate ventilation and illumination in the generator room



Figure 2.48 - No power supply to lighting and exhaust fan in LT panels and generator area

10.2 Working clearance

This section refers to the electrical switch room and generator room.

During our inspection, the working clearances surrounding key electrical equipment were assessed.

It was observed that the working clearance was inadequate in multiple areas. Equipment was placed too close to walls or other machines, making it difficult for personnel to access for routine maintenance or emergency interventions. Additionally, some pathways were obstructed, restricting movement and potentially delaying response times during an emergency. This issue was specifically observed with the 220kVA generator, the LT panels, and the changeover panel in the generator room.

Proper working clearance is crucial for ensuring life and fire safety. Adequate clearance allows personnel to safely access and maintain equipment, thereby reducing the risk of accidents or injuries caused by cramped conditions. Insufficient working clearance increases the risk of equipment failure due to limited maintenance access, which can lead to fires or other hazardous situations. F-202603-064258

Timeline	Remediation Plan
Medium Term CAP-0075597	Reorganize the equipment layout to ensure adequate clearance, remove obstructions in pathways, and implement clear floor markings and signage to designate working areas and pathways.
Longer Term CAP-0075598	Conduct regular workspace audits to maintain clearances, update maintenance protocols to include clearance checks, provide personnel training on maintaining clearances, and consider installing adjustable equipment to accommodate flexible positioning.



Figure 2.49 - Inadequate working clearance for 220kVA generator



Figure 2.50 - Inadequate working clearance for LT panels and changeover panel installed in generator room

10.3 Battery

During the inspection of the battery system, the battery cells, connections, and enclosures were thoroughly examined.

It was observed that generator battery terminals were exposed and not properly insulated, increasing the risk of electrical shock or short circuits.

Proper monthly and annual maintenance of the battery system is crucial for fire safety. Spilled fluid can cause corrosion and fires, while bare terminals increase the risk of electrical shocks and short circuits, potentially igniting fires or explosions. Non-compliant battery placement can obstruct ventilation, leading to hazardous gas buildup and fire risks. Addressing these issues is essential to prevent fires and ensure battery system safety.F-202603-064257

Timeline	Remediation Plan
Medium Term CAP-0075595	Insulate exposed battery terminals using appropriate insulating materials to prevent electrical shock and short circuits. Conduct a thorough inspection of all connections to ensure that there are no immediate risks of exposure.
Longer Term CAP-0075596	Establish a routine inspection and maintenance protocol to check and maintain proper insulation of all battery terminals regularly.



Figure 2.51 - Exposed terminals observed on the 100kVA generator batteries

10.4 Back-up power supply

During the assessment of the backup power supply system, an inspection was conducted on the generators, battery backups, and auxiliary power sources.

It was observed that the backup supply generators exhibited signs of inadequate maintenance, including worn components and outdated fuel filters.

Maintaining a reliable backup power supply system is crucial for ensuring continuous operations and safety during power outages or emergencies. Inadequate maintenance of generators can lead to failures during critical moments, jeopardizing essential services and potentially causing disruptions or safety hazards.F-202603-064256

Timeline	Remediation Plan
Medium Term CAP-0075592	Perform comprehensive generator maintenance, including rectification of oil leakages, treatment and protection of corroded components, replacement of worn parts and fuel filters, and a thorough inspection to address any additional safety and performance deficiencies.
Longer Term CAP-0075593	Establish a regular maintenance schedule to ensure all generators are routinely serviced, including timely replacement of components and fuel filters.



Figure 2.52 - Backup supply generator showing corrosion and inadequate maintenance



Figure 2.53 - Backup supply generator showing corrosion and oil leakages



Figure 2.54 - Backup supply generator showing inadequate maintenance of components

11. Cabling & Wiring

11.1 Cable support and protection

This section refers to the entire facility.

During the inspection of the entire facility, a visual survey of the cable containment system was conducted.

The following observations were noted:

i. Proper supports were not provided for cables; they were laid directly on the ground and hung along the walls. Additionally, several locations had open and exposed electrical connections. Inadequate cable support results in sagging or stress on cables, which can lead to damage or failure. This issue was specifically observed at the main gate entrance and in the generator room area.

ii. Cable trays and conduits were found rusted and inadequately supported at multiple locations, including at the main gate entrance, in the cutting, stitching, and store areas of building 3, and on the first floor of building 6. Such deterioration and insufficient support can lead to cable damage, overheating, and potential electrical faults or fire hazards.F-202603-064234

Timeline	Remediation Plan
Medium Term CAP-0075731	i. Secure cables with appropriate supports and insulate any exposed connections to prevent sagging, stress, and potential hazards. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support and 526.5 for proper terminations).
Medium Term CAP-0075732	ii. Replace or refurbish rusted cable trays with corrosion-resistant trays and ensure proper secure mounting according to BS 7671 18th Edition + A2 (2022) Regulation 522.8.5 for mechanical support.
Longer Term CAP-0075733	Implement a comprehensive cable management system with regular inspections to ensure cables remain properly supported and the cable trays are corrosion-free.



Figure 2.55 - Inadequate support for the cable tray in the cutting area of building 3



Figure 2.56 - Inadequate support for the cable tray on the first floor of building 6



Figure 2.57 - Inadequately supported cables and corroded cable tray at the main gate entrance



Figure 2.58 - Electrical cables laying on the ground in the generator area



Figure 2.59 - Inadequate support for the cable conduit in the stitching area of building 3

11.2 Cable trench and channel cleanliness

This is related to the whole facility.

During the inspection of the facility, it was observed that the cleanliness of the cable trunking system had not been maintained. Numerous trunking systems were found to be filled with a significant amount of fluff and dust, which poses a serious fire risk. This issue was specifically observed with the cable trays of buildings 4 and 9.

Accumulated dust and fluff can ignite when exposed to sparks or high temperatures, potentially leading to a fire hazard. Maintaining a clean cable trunking system is essential to minimize fire risks associated with accumulated debris.F-202603-064237

Timeline	Remediation Plan
Medium Term CAP-0075769	Clean all cable trunking systems to remove accumulated dust and fluff, ensuring that any potential fire hazards are mitigated.
Longer Term CAP-0075770	Implement a regular maintenance schedule that includes routine inspections and cleaning of the cable trunking systems to prevent future accumulation of dust and fluff, thereby maintaining a safe environment and complying with safety standards.



Figure 2.60 - Lint and dust inside the cable tray on the first floor of building 9



Figure 2.61 - Lint and dust inside the cable tray of building 4

11.3 Splicing of cables

This relates to the whole facility.

During the inspection, the cable installation was visually surveyed throughout the facility.

The following observations were noted:

i. Cables terminations within panels and switchgears were made without cable lugs, resulting in broken strands and loose connections. This presents a significant fire risk due to the potential for electrical arcing and overheating, which can ignite nearby materials and cause a fire. Furthermore, loose connections can cause voltage drops and equipment malfunctions, posing a life safety hazard. This issue was observed in most of the distribution boards, including the gate office distribution board of building 1, the stitching distribution board of building 3, and the gloves distribution board of building 2.

ii. PVC tape joints were found in several locations - including within distribution boards - where cables were either damaged or two cables were being joined together. This method of jointing is unreliable and can deteriorate over time, increasing the risk of electrical shorts and fires. The adhesive on PVC tape can degrade, leading to exposed conductors that can spark and ignite surrounding materials, presenting both fire and life safety risks. This issue was specifically observed at the generator cable connections and at the water pump connection near the main gate.

iii. Machine power wires were inserted directly into power receptacles without a power plug. This practice is highly dangerous as it can lead to loose connections, electrical shocks, and short circuits. Direct insertion bypasses proper safety mechanisms, significantly increasing the risk of fire and electrical hazards, thereby endangering life safety. This issue was specifically observed on the first floor of building 3 and on the ground floor of building 4.

iv. Non-compliant and low-quality PVC connectors used in the cable trays, which could pose a hazard of a potential short circuit due to their poor insulation. Observed on the first floor of building 9.

v. Open, exposed, and idle cable connections were found inside distribution boards, posing a risk of electric shock, short circuit, and fire. This issue was specifically observed in the gate office distribution board of building 1 and the gloves distribution board of building 2.

vi. Machine plugs were installed directly into incompatible sockets, creating unsafe electrical connections and increasing the risk of overheating, arcing, and fire. This issue was specifically observed in the cutting area of building 3, on the first floor of building 6, and on the first floor of building 9.

For any future electrical work, the termination of stranded cables must be done with cable lugs, and PVC tape joints shall not be utilized. Proper termination ensures secure and reliable connections, reducing the risk of electrical failures and enhancing overall safety. F-202603-064235

Timeline	Remediation Plan
Medium Term CAP-0075743	i. Install cable lugs on all improperly terminated cables within panels and switchgears to secure connections and prevent breaking strands and loose connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022).
Medium Term CAP-0075745	ii. Replace all PVC tape joints with proper terminal blocks or connectors to ensure secure and stable connections in accordance with Regulation 526.1 of BS7671 18th Edition + A2 (2022). Replace all damaged cabling with new wiring.
Medium Term CAP-0075746	iii. Replace the directly inserted power wires with proper power plugs to ensure safe and secure connections.
Medium Term CAP-0075747	iv. Replace the low-quality PVC connectors with standard quality terminal blocks in accordance with BS7671 18th Edition + A2 (2022).
Medium Term CAP-0075752	v. Remove unused cables or properly terminate and insulate open and exposed connections to eliminate the risk of sparks and potential fires. (Reference: BS 7671 18th Edition + A2 (2022) Regulation 416.1 for basic insulation of live parts).
Medium Term CAP-0075753	vi. Stop the use of unsafe plug-in arrangements, provide compatible dedicated sockets for machines (preferably industrial sockets), and enforce safe electrical connection practices.
Longer Term CAP-0075754	i. Implement a standard operating procedure (SOP) requiring the use of cable lugs for all future cable terminations. Regularly train and audit personnel to ensure compliance with this procedure, thus maintaining safe and reliable electrical connections.
Longer Term CAP-0075755	ii. Establish a policy prohibiting the use of PVC tape for electrical jointing. Regularly inspect and maintain electrical installations to ensure compliance, and provide continuous training for staff on proper jointing techniques.
Longer Term CAP-0075756	iii. Develop and enforce a facility-wide electrical safety protocol that mandates the use of power plugs for all equipment. Conduct regular safety audits and provide training sessions to staff to reinforce the importance of using proper power plugs and adherence to electrical safety standards.

Longer Term CAP-0075757	iv. Establish a plan to replace all low-quality PVC connectors with compliant electrical terminal blocks and integrate this into regular maintenance to ensure ongoing safety and compliance.
Longer Term CAP-0075758	v. Establish routine inspections to ensure all electrical terminations remain properly insulated and protected from exposure, especially in high-risk operational areas.
Longer Term CAP-0075759	vi. Conduct regular safety audits and provide training sessions to staff to reinforce the importance of using proper power plugs and adherence to electrical safety standards.



Figure 2.62 - Improperly spliced generator cable joints covered with PVC tape



Figure 2.63 - Improper plug-in connections observed for the stitching machines on the first floor of building 9



Figure 2.64 - Low quality PVC connector found in the cable tray on the first floor of building 9



Figure 2.65 - Improper plug-in connection observed for the cutting machines in building 3



Figure 2.66 - Open connections and cable terminations without lugs found in the gate office distribution board, building 1



Figure 2.67 - Machine power wires inserted directly into power receptacles without a power plug, building 3



Figure 2.68 - Open and exposed connections found in the building 4

11.4 Colour coding

This relates to the whole facility.

During the inspection of several distribution boards, deviations from standard cable colour coding practices were identified. Such deviations pose significant risks, as they can lead to the misinterpretation and misidentification of cables. This misidentification can result in incorrect connections, potentially causing unsafe working conditions and increasing the likelihood of electrical accidents, such as shocks or fires. This issue was observed in all distribution boards across the facility, including those in buildings 2 and 3.

To mitigate these risks, it is imperative that all future electrical work strictly adheres to standardized colour coding as prescribed by BS 7671. This will ensure accurate identification of conductors, thereby enhancing overall safety and operational efficiency. F-202603-064236

Timeline	Remediation Plan
Medium Term CAP-0075767	Conduct a comprehensive audit of all distribution boards to identify and correct any deviations from the standard cable colour coding. This involves re-labelling or re-insulating the cables with the appropriate colour codes as specified in BS 7671. Specifically, adhere to BS 7671 Regulation 514.3.1, which mandates the correct identification of conductors.
Longer Term CAP-0075768	Develop and enforce a strict protocol for cable colour coding to be followed for all future installations and maintenance activities. This protocol should include regular training sessions for electricians and maintenance personnel to keep them updated on the latest standards and practices. Additionally, periodic audits should be conducted to ensure ongoing compliance with the standardized colour coding system, thereby preventing future deviations and ensuring sustained safety and operational efficiency.



Figure 2.69 - Incorrect color coding in the cutting distribution board of building 3



Figure 2.70 - Incorrect color coding in the stitching distribution board of building 3



Figure 2.71 - Incorrect color coding in the gloves distribution board of building 2

11.5 Fixtures

This relates to the whole facility.

During the inspection, the electrical fixtures throughout the facility were visually surveyed.

The following observations were noted:

i. Power receptacles mounted on combustible wooden ply found on the first floor of building 6. Installing receptacles on combustible materials like wooden ply increases the risk of fire if a short circuit or overheating occurs. Combustible materials can easily ignite, leading to potential fire hazards and endangering life safety.

ii. Burnt power receptacles found on the ground floor of building 8: Damaged receptacles with exposed parts can lead to loose connections and increased electrical resistance. This condition can cause overheating, potentially igniting surrounding materials and increasing the risk of electrical fires, posing a serious life safety threat.

iii. Lighting tube rods and LED lights were found suspended directly from cables, and several fixtures were broken due to a lack of proper support, posing a risk of cable strain, electrical faults, or fire hazards. This issue was observed across multiple buildings, including at the main gate entrance, the generator room, and buildings 3 and 9.F-202603-064238

Timeline	Remediation Plan
Medium Term CAP-0075771	i. Replace combustible wooden ply backings with non-combustible materials to reduce fire risk.
Medium Term CAP-0075772	ii. Replace or repair burnt and exposed power receptacles immediately to restore safe and secure connections.
Medium Term CAP-0075773	iii. Install appropriate lighting fixtures with secure mounting provisions to support bulbs and LED tube lights independently, without relying on electrical cables.
Longer Term CAP-0075774	i. Establish a facility-wide standard to use only non-combustible materials for mounting electrical components, and conduct regular audits for compliance.

Longer Term CAP-0075775	ii. Implement a preventive maintenance program that includes periodic checks and timely replacement of damaged electrical components to maintain safety standards.
Longer Term CAP-0075776	iii. Establish and implement a facility-wide standard for the proper installation of all lighting fixtures using durable supports, to prevent stress on electrical wiring and ensure long-term safety and reliability.



Figure 2.72 - Burnt out power receptable found on the ground floor of building 8



Figure 2.73 - Inadequately supported and broken lighting fixture in leather store of building 9



Figure 2.74 - Inadequate support for light fixtures at main gate entrance



Figure 2.76 - Inadequately supported tube rods and light bulbs in cutting area of building 3



Figure 2.75 - Power receptacle installed with wooden backing in the on the first floor of building 6

12. Assessment Limitations and Assumptions

The information in this report was obtained during an inspection of the facility and based on information provided by factory management. It has not been possible to provide independent verification for all the information and data collected, and therefore, the Accord cannot accept general responsibility for omissions or errors arising from inaccuracies in this report from the information obtained.

The findings and recommendations in this report are not intended to imply, guarantee, ensure or warrant compliance with any government regulations. Additionally, the results do not imply in any way that compliance with the findings or recommendations as stated in this report will eliminate all hazards, risks or exposures or that hazards, risks or exposures not referred to in this report do not exist. Compliance with the findings and recommendations stated in this report does not relieve the factory owner from obligation to comply with specific project requirements, industry standards, or the provisions of any local government regulations.

Except as otherwise noted within this report, other systems or services were not viewed or tested during the assessment and are therefore similarly excluded from this report. We have not assessed any parts of the structure / building fabric / services which are covered, unexposed or inaccessible and we are therefore unable to report that any such part of the property is free from defect.

The observations in this report are based on the engineering judgement of the engineer at the time of the inspection.